

CAS Notes and Presentation Prep

CAS Basics

CAS means Counterfactual Attention Swap.

That sounds scary, but it is actually simple.

Break it into 3 parts:

Counterfactual means: "what if this part were different?"

Attention means: "which part of the image seems important to the model?"

Swap means: "replace that important part with another patch"

So CAS is basically this:

find the important region in an MRI

change that region

prediction changes

run the model again

see if the model tells us whether the model was really relying on that region.

Very Simple Example

Suppose the model sees an MRI and says: glioma.

Now we find the area the model seems to focus on.

Then we replace that area with a patch from another MRI.

Now we test again. If the model changes from glioma to meningioma, or its confidence drops a lot, then that region was important.

So CAS helps answer:

Why did the model predict this?

Not just:

Was the model right or wrong?

In Your Project

Your project has 3 parts:

1. CNN: one model for brain MRI classification

2. Vision Transformer (ViT): another model for the same task

3. Vision: a method to compare how sensitive these models are to an important image region

In your notebook, CAS works like this:

use CNN saliency to find the most important region

swap or blend that region with a patch from another-class MRI

give the changed image to both CNN and ViT

compare how their predictions change

Later, you also discussed doing ViT-guided CAS, where ViT chooses the important region.

What Saliency Means

Saliency is just a heatmap of importance.

It shows which pixels influenced the prediction more.

Brighter area = more important.

So when you say "we used CNN saliency", you mean:

we used the CNN to find which MRI

CAS Notes and Presentation Prep

region looked most important for its decision.

Why CAS Is Useful

Accuracy tells us how often the model is correct.

CAS tells us:

depends on:
what region the model
is sensitive to, and whether
whether one model is more

So CAS adds interpretability.

That means helping us understand model behavior.

What Happened in Your Results

Your important finding was something like:

CNN stayed stable after the swap
ViT changed much more after the swap

That means:

CNN was less affected by the region change
ViT depended more strongly on
that swapped region

So your conclusion is:

CNN and ViT do not rely on image evidence in exactly the same way.

One-Line Definition for Presentation

Counterfactual Attention Swap is a method where we identify an important image region, modify or swap it, and observe how the model's prediction changes.

Super Easy Speaking Version

CAS is basically a what-if test. We first find the MRI region that looks important to the model. Then we change that region and check the prediction again. If the prediction changes a lot, that region was important.

Presentation Flow

1. Problem: We classify brain MRI images into tumor categories.
2. Models: We used CNN and Vision Transformer.
3. Comparison: We compared their accuracy and confusion matrices.
4. CAS idea: We wanted to understand not just which model is better, but why.
5. CAS method: Find important region, swap that region, re-test both models.
6. Result: CNN and ViT reacted differently, so they use image evidence differently.

What To Say Slide By Slide

Slide: Project

We worked on brain MRI classification. We used two models, CNN and Vision Transformer, and then added CAS to make the comparison more interesting.

Slide: CNN vs ViT

CNN and ViT were both trained on the same dataset. We compared them using test accuracy, validation accuracy, and confusion matrices.

CAS Notes and Presentation Prep

Slide: What is CAS

CAS stands for Counterfactual Attention Swap. It is a way to test whether an important MRI region is actually influencing the model's decision.

Slide: How CAS works

First, we find the important region using saliency. Then we swap or modify that region using a patch from another MRI. Then we run the models again and compare the prediction changes.

Slide: CAS result

In our experiment, the Vision Transformer was more sensitive to the swapped region than the CNN. This suggests the two architectures rely on visual evidence differently.

Very Important Viva Answers

Why did you use CAS?

We used CAS to understand model behavior, not just model accuracy.

Why use saliency?

Saliency helps us estimate which image region is important for the prediction.

What does prediction change mean?

It means the model depended on that region. A bigger change means higher dependence.

Why compare CNN and ViT with CAS?

Because both may classify correctly, but CAS helps show whether they make decisions using the same visual evidence or different evidence.

Short Script You Can Memorize

CAS stands for Counterfactual Attention Swap. In simple terms, it is a what-if experiment. We first find the region of the MRI that seems important to the model. Then we modify or swap that region and test the model again. If the prediction changes a lot, it means that region was important. We used this to compare CNN and Vision Transformer, and we found that they respond differently to the same attention-guided change.

How is this idea good and different?

It is good and different because most student projects stop at training a model, showing accuracy, and comparing results. Your project goes one step further. Instead of only asking which model is more accurate, you also ask why the model is making that decision, which part of the MRI is important, and whether CNN and ViT rely on the same region or different regions.

Why it is good:

Not just accuracy.
Compares two architectures more deeply

Adds interpretability

Good for healthcare context

Why it is different:

Most teams may do CNN vs ViT plus an accuracy table and confusion matrix. Your project also adds attention-based analysis, counterfactual testing, and sensitivity comparison.

CAS Notes and Presentation Prep

Simple way to say it:

Our project is different because we did not stop at model comparison. We also used CAS to understand how each model reacts when an important MRI region is changed.

Best one-line answer:

It is a good idea because it compares both performance and interpretability, not just accuracy.

Even simpler:

Most projects ask which model is better. Our project also asks why the model is behaving that way.

How is it masking and making CNN behave like ViT and ViT behave like CNN?

It is not making CNN behave like ViT or ViT behave like CNN.

CAS only finds an important region, changes that region, gives the changed image to both models, and compares their reactions.

So CAS is testing the models, not turning one model into the other.

What it is doing:

same image
observe who changes more
same swapped region
with models seeing the changed version

We send CNN features into ViT
converting ViT into CNN
Not converting CNN into ViT
Not forcing one model to think like the other

Correct sentence to say:

CAS does not make CNN behave like ViT. It compares how both models react when the same important image region is changed.

Can we make CNN behave like ViT and ViT behave like CNN?

In theory, a little, but not with your current CAS code.

That would need advanced ideas like knowledge distillation, attention transfer, feature alignment, or hybrid CNN-ViT architectures.

Best class answer:

Currently, CAS compares the behavior of CNN and ViT under the same attention-guided perturbation. It does not force one model to behave like the other. Making them behave more similarly would require methods like attention transfer, distillation, or hybrid architectures.

For this project, we are sticking to CAS.